Enriching Cultural Heritage through Semantic Annotation: A Review of Methods, Tools, and Collaborative Spaces

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Introduction

This paper presents a comprehensive review of semantic annotation practices within Digital Humanities (DH) projects. As the demand for enriched, interoperable cultural data increases, semantic annotation and Linked Open Data (LOD) have become central in connecting heritage materials across digital platforms. Focusing on DH projects as the unit of analysis, we examine current methodologies, tools, and frameworks for annotation, emphasizing their role in research workflows. We propose a multidimensional classification schema to support scholars and institutions in evaluating, comparing, and adopting annotation strategies. The paper concludes with recommendations for fostering shared frameworks and advancing semantic practices in DH.

What is an Annotation?

Annotations in the humanities and cultural heritage fields have been recognized as a "scholarly primitive" (Unsworth 2000). The digital paradigm introduces significant differences in the production, management, and use of annotations. Within the DH domain, annotations have evolved to become increasingly employed as essential "tools for research" and are emerging as valuable research data in their own right (Walkowski 2016). The purposes of annotation vary widely based on factors such as the nature of the annotated object and user requirements, encompassing everything from informal commentary to structured relationships between resources.

From a general point of view, annotating is the act of associating information together (Sanderson et al. 2013). An annotation is typically structured into three components (Boot 2009): (1) the *annotatum* , or target, the object being annotated; (2) the *annotans* , or body, which constitutes the content of the annotation; and (3) the *annotator* , who establishes the connection between body and target.

In the context of the Semantic Web, the term *semantic annotation* varies in meaning depending on the field of application (Sharma/Jain 2021). In linguistics, it involves tagging text with metadata to capture linguistic features, often using automated techniques (e.g., part-of-speech tagging). In digital scholarly practices, the term has been associated with the expression *scholarly annotation* (Pedretti et al. 2024), shifting the focus from annotation intended as metadata (Agosti 2004) to annotation as a hermeneutical process conducted by scholars (Bradley 2012). In this paper, we define *semantic annotation* as the practice of using URIs to identify entities and their relationships, forming RDF triples unambiguous for machines (Di Donato et al. 2013). This definition draws a clear boundary from analogic annotations expressed in natural language—often described in the literature as “ontological” versus “informal” annotations (Oren et al. 2006)—yet it also invites critical reflection on what is meant by “semantics” in annotation practices. While the use of URIs is central to the principles of LOD, it may fall short of capturing the interpretive dimension emphasized in DH scholarship. Furthermore, digital technologies have significantly expanded the range of contexts in which annotations are used, highlighting the need for more nuanced reflection on the tools and methodologies adopted within the DH community (Walkowski 2016). Despite these complexities, the URI-based definition offers a functional basis for analyzing DH projects that employ semantic technologies.

Methodology

Our survey examined annotation projects across various types of digitized data, including text, multimedia, and 3D objects. Projects were selected with a particular interest in the role of end users. As a result, systematic annotation relying solely on machine-learning techniques was exclu-

ded. Instead, the focus was on projects where annotations are created manually or under human supervision in cases where semi-automatic methods are employed.

For the selection process, we adopted a multifocal approach (Windhager 2019) by identifying domain-specific journals and conference databases, which were queried using predefined keywords to retrieve relevant publications. An empirical approach was then employed, exploring references and related projects cited within the retrieved works.

Regarding the domain of interest, our focus primarily encompassed the area of DH, including fields such as Digital Scholarship, Digital Libraries, Digital Art History, Applied Computing to Arts and Humanities, and Digital Cultural Heritage. Additionally, we considered the practice of annotation within the Semantic Web. Relevant journals included *Digital Humanities Quarterly* (DHQ), *Journal on Computing and Cultural Heritage* (JOCCH), and *Digital Scholarship in the Humanities* (DSH). Furthermore, we explored results from scholarly organizations in the DH field, such as the International Image Interoperability Framework[1] (IIIF) Community, which is particularly active in advancing digital image annotation as a central task.

The final dataset[2] comprised 30 projects, of which 17 have an associated research paper. Additionally, 8 projects were described in research papers but lacked an implemented prototype. The remaining 5 projects consisted of web-based applications or prototypes without any associated research publication.

A Classification Schema for Annotation Systems

To investigate DH projects that implement semantic annotations, we developed a multidimensional classification with a focus on productive processes (i.e. tools, activities, and users) as means for generating new data. The resulting schema (Figure 1) offers a structured yet flexible foundation for further discussion and refinement.

Ranging from structural aspects of annotation itself, up to the analysis of the end users of annotation tools, the schema follows three main axes: *data* , *tasks* and *users* (Windhager 2019).

Additional dimensions beyond the classification schema include discoverability, interoperability, and reusability of annotation data, along with considerations of data storage, provider reliability, licensing, and security. Crucial aspects like uncertainty, provenance, and versioning are also vital for ensuring the transparency, reliability, and long-term usability of semantic annotations.

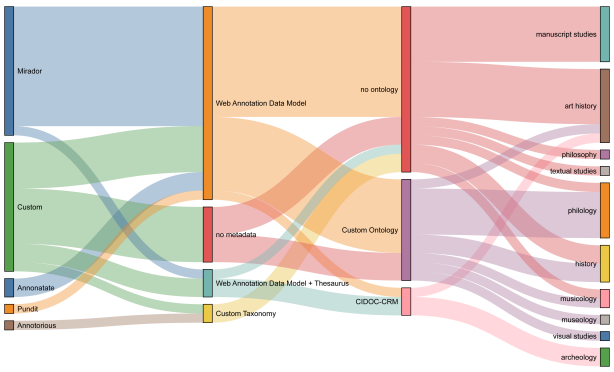

Figure 1. Multidimensional Classification Schema for Annotation Systems

Data

Data refers both to the structure or format of the annotated entity (i.e., text, image, etc.) and the formal conceptualization of the annotation itself (i.e., metadata, vocabularies and ontologies). In particular, annotations can be classified into formal or structured (machine-readable), informal or unstructured (human-readable), and ontological (both human- and machine-readable) (Oren et al. 2006). For this study, we adopted a dichotomy that distinguishes between annotations based on generic metadata, which may include both informal and formal elements, and those grounded in ontologies, which are explicitly structured to support semantic reasoning.

Finally, the production of annotations involves processes and techniques that span a spectrum from manual control to full automation. Annotations can be created manually, as in the interpretative work of a domain expert; semi-automatically, with computational tools aiding the process but requiring human validation or refinement; and automatically, with algorithms or machine learning models.

Tasks

Annotation purposes can be related to descriptive activities (i.e., providing contextual information), analytical activities (i.e., adding insights or interpretations) and relational activities (i.e., linking elements or entities). For categorization, we derived some concepts from the classification proposed by the Taxonomy of Digital Research Activities in the Humanities[3] (TADIRAH).

In particular, we selected *tagging* , *commenting* , *transcribing/translating* , *associating* , and *highlighting* .

On the other hand, tools are frameworks or applications employed to facilitate the annotation process. These tools provide the technical means for creating, managing, and linking annotations to data, often tailored to the needs of a project. Tools can be standardized, such as Mirador[4] , which leverages the IIIF for annotating image-based resources, or custom-built, designed for unique requirements. The choice of tool significantly impacts the interoperability and scalability of the annotation workflow, making it a critical consideration in semantic annotation projects.

Users

From a user need perspective, we analysed annotation systems by intended audience (Chiang 2010): private (e.g., individual researchers), collaborative (e.g., research groups), or meant to be public. In the first case, the annotation system should support the creation of private annotations or allow a user to publish them in a later stage. A collaborative annotation system should support an environment in which different users can operate on the same target annotation. Lastly, public annotations are visible to everyone and can be interacted with to a certain extent.

The target users for semantic annotation in the DH range from domain experts, such as humanities scholars, to casual users seeking meaningful insights. Between these extremes are curation managers, who facilitate data enrichment, and special-purpose users, who employ annotation tools for didactic purposes. For casual users, who may lack extensive knowledge of the data, accessible interfaces and workflows remain essential.

Results

Our analysis[5] of target, purposes, and audience categories (Figure 2) shows that most annotation tools prioritize commenting, with no support for highlighting. This suggests a general preference for interpretive and descriptive uses of annotations.

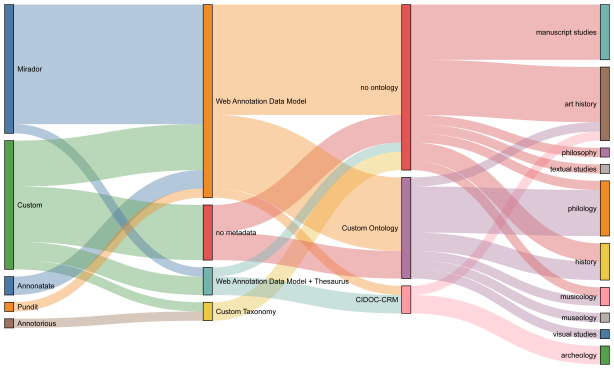

Figure 2. Visualization of Target, Purposes and Audience within Annotation Projects.

Recent trends[6] indicate an increasing adoption of Mirador and the Web Annotation Data Model (WADM), reflecting a shift toward standardized, reusable solutions. In contrast, ontology-based models remain less common, likely due to the complexity of implementing a full semantic stack (Figure 3).

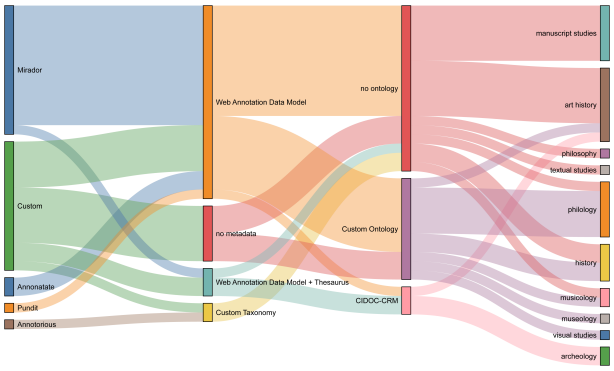

Figure 3. Visualisation of Tools, Metadata and Ontology used within Annotation Projects.

A partial correlation was observed between the application domain[7] and the choice of technologies (Figure 4). Projects focused on art history and image annotation tend to adopt visual and IIIF-compliant tools (e.g., Mirador), whereas text-oriented projects (e.g., philology or history) often employ custom-developed frameworks.

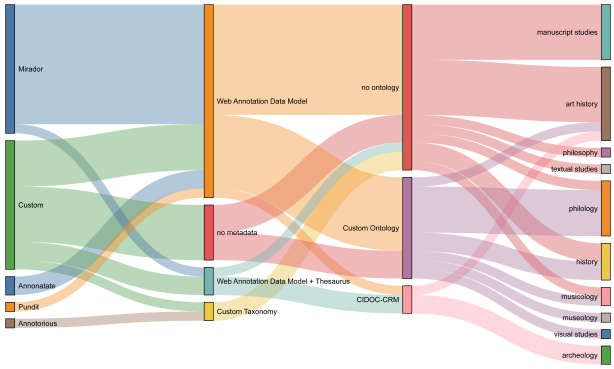

Figure 4. Visualisation of Tools, Metadata and Ontology adopted by different Domains.

Conclusions

By explicitly organizing the multidimensional aspects of annotation practices, the classification schema could help facilitate both theoretical reflection on current trends and informed decision-making for project planning within the DH community, particularly for scholars, project designers, and tool developers who need to compare, evaluate, or select annotation systems based on specific features.

Overall, we analyzed a dataset of 30 projects, finding that only 33% included a user study. Evaluation is often absent, raising concerns about validation and usability, highlighting the need for a stronger emphasis on evaluation in DH projects.

Information visualization remains underexplored, leading to oversimplified presentations of annotations. Additionally, limited attention to provenance in formal models for annotation leaves annotations subject to varying degrees of epistemic and ontological uncertainty.

In conclusion, thoughtful design and standardized workflows are essential to enhance the robustness and usability of semantic annotation practices in the DH.

Notes

1. https://iiif.io/ .
2. For further details, the dataset and additional materials are accessible at the following links: https://github.com/friendlynihilist/sem-annotation-survey and https://doi.org/10.5281/zenodo.14316772 .
3. https://vocabs.dariah.eu/tadirah/en/page/annotating
4. https://projectmirador.org/
5. In-depth visualizations resulting from the dataset analysis are accessible at the following link: https://friendlynihilist.github.io/sem-annotation-survey/ .
6. Identifying precise temporal trends is challenging due to the often unclear lifecycle of DH projects, making it difficult to distinguish between discontinued, inactive, or ongoing initiatives. For visualizations see https://friendlynihilist.github.io/sem-annotation-survey/ .
7. Given the small size of the dataset, the domains' identification was based on a close analysis of the documentation provided. As the dataset expands, we plan to introduce automated methods such as keywords extraction from publications abstracts to support domain classification at scale.

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